

Meridian Parts

European service-hub expansion

Board decision brief | 2025 operating evidence, option economics and a gated 90-day plan

Prepared for the synthetic client. Source room collected 27 September 2026; analysis cutoff: 31 January 2026 inclusive.

Executive answer

Fund a staged CZE + ESP service-hub path, subject to operating gates. The option is the highest base-case annual incremental contribution among all feasible singles and pairs under the client's no-synergy policy: **€198,099 per year on €225,000 year-zero capex and 5 FTE.**

Measure	Low	Base	High	Defined stress
Annual incremental contribution	€64,619	€198,099	€331,580	€32,180
Simple payback	3.48 yr	1.14 yr	0.68 yr	6.99 yr

Decision rule and caveat: the recommendation maximizes modeled base annual contribution, not risk-adjusted NPV. The model is scenario arithmetic built from synthetic client inputs; it is not statistically proven, causal or certain. Stage the second site so the board can stop after measured evidence.

Why this option

CZE combines the strongest modeled hub economics (EUR95k capex, 2 FTE, EUR2.10/unit saving) with a positive low-volume case; ESP adds the highest baseline contribution in the six-market set and remains positive under the defined joint stress. Together they use 5 FTE and EUR225k—within both limits—and beat the strongest base-case alternative POL+ESP by EUR11.8k annual contribution while using one fewer FTE and EUR15k less capex.

Board approval requested

Approve a two-stage envelope of EUR225k capex and 5 FTE, release CZE first, and authorize ESP only when the gates in the implementation plan are met. If the board requires a one-site-only commitment, CZE is the better first site on modeled base contribution, low-volume contribution and capex/FTE efficiency.

1. What the 2025 evidence says

The accounting view is built from shipped orders in 2025, with returns received through 31 January 2026. It uses the source-room dictionary: one order is one row after highest revision selection; identical repeats are collapsed; cancelled/test/future orders are excluded; valid zero-price shipments are retained; distinct return IDs are additive; orphan returns are quarantined.

2025 network measure	Value
Shipped orders	1,254
Shipped units	77,436
Gross sales	€4,464,622
Refunds	€80,831
Net sales	€4,383,792
Net COGS	€2,246,919
Fulfillment	€106,862
Contribution / margin	€2,030,010 / 46.3%
Returned units	1,081 (1.4% of units)

Country diagnosis

Country	Orders	Units	Net sales	Contribution	Margin	Returned units
DEU	209	12,153	€625,792	€262,366	41.9%	73
FRA	209	11,880	€632,231	€276,271	43.7%	144
NLD	209	12,576	€697,733	€315,644	45.2%	144
POL	209	13,163	€761,890	€356,494	46.8%	180
CZE	209	13,733	€815,307	€397,335	48.7%	288
ESP	209	13,931	€850,839	€421,900	49.6%	252



ESP and CZE lead modeled baseline contribution and margin. This is a starting point for hub economics—not proof that local fulfillment caused the difference. Returned-unit rate is highest in CZE (2.1%) and ESP (1.8%); the implementation KPI plan therefore puts return rate and restock recovery beside volume and savings.

Booked revenue, cash and contribution

Gross sales are the booked shipment value before explicit refunds. Net sales subtract qualifying refunds known by the cutoff; this is not a cash-receipts measure because customer payment and refund settlement timing are not provided.

Contribution subtracts net COGS and actual nonrefundable fulfillment from net sales; it excludes hub capex, hub fixed cost, taxes, financing and corporate overhead. The ECB quote is an analytical translation rate, not proof of the actual cash conversion rate.

2. Market context: useful context, not demand proof

The archived World Bank responses provide comparable 2022–2024 population and GDP per capita (current US\$). Population is a scale indicator and GDP per capita is an economic-context indicator; neither is a direct proxy for Meridian Parts replacement-assembly demand.

Country	Population 2022	Population 2024	Change	GDP pc 2022	GDP pc 2024	Change
DEU	83,177,813	83,516,593	0.4%	US\$50,507	US\$56,104	11.1%
FRA	68,184,457	68,551,653	0.5%	US\$40,989	US\$46,103	12.5%
NLD	17,700,982	17,993,485	1.7%	US\$59,123	US\$67,465	14.1%
POL	36,821,749	36,559,233	-0.7%	US\$18,891	US\$25,104	32.9%
CZE	10,672,118	10,905,028	2.2%	US\$28,282	US\$31,828	12.5%
ESP	47,786,102	48,848,840	2.2%	US\$30,319	US\$35,327	16.5%

Population grew in CZE and ESP by 2.2% from 2022 to 2024, while POL declined 0.7%. Current-US-dollar GDP per capita rose in all six markets, with the largest percentage change in POL; exchange rates and nominal-price effects are embedded in that observation. Source: archived World Bank API responses in `evidence/raw/population.json` and `gdp-per-capita.json`, vintage last updated 13 July 2026.

FX method

PLN and CZK prices and refunds were divided by the original sale-month arithmetic mean of available ECB business-day reference quotes in local currency per EUR. The analysis does not invert the quote or use today's rate. 2025 mean ranges were PLN 4.1722–4.2658/EUR and CZK 24.2341–25.1633/EUR across the twelve months; EUR transactions use 1.0000. Full monthly rates and observation counts are in the workbook and `analysis/outputs/fx_monthly.csv`.

3. Hub economics and choice

The policy defines annual incremental contribution as $C \times u + U \times (1+u) \times s - F$, where C is 2025 contribution, U shipped units, u is the low/base/high uplift, s is per-unit saving and F is recurring fixed cost. Capex is kept separate and used only for simple undiscounted payback. Stress applies a 3% gross-sales refund shock plus a 10% net-sales depreciation shock for PLN/CZK markets, then uses 25% volume uplift. Pairs add country figures without synergy.

Top feasible pairs by base case

Alternative	Low	Base	High	Stress	Capex	FTE	Base payback
CZE+ESP	€64,619	€198,099	€331,580	€32,180	€225,000	5	1.14 yr
POL+ESP	€58,562	€186,329	€314,096	€29,264	€240,000	6	1.29 yr
POL+CZE	€56,856	€178,995	€301,134	-€78,570	€205,000	5	1.15 yr
NLD+ESP	€43,994	€166,931	€289,869	€107,630	€270,000	6	1.62 yr
NLD+CZE	€42,289	€159,598	€276,907	-€203	€235,000	5	1.47 yr

CZE+ESP is the recommended pair. POL+ESP is the strongest alternative on base contribution, but it is EUR11.8k lower, costs EUR15k more capex and requires one additional FTE; its defined stress remains positive but less resilient than CZE+ESP. POL+CZE is cheaper but its joint stress is negative because both markets take the PLN/CZK shock. Full singles and all feasible/infeasible pair combinations are in the workbook's All options and Pair scenarios tabs.

What would change the decision

The critical unknown is realized hub uplift and saving—not population or nominal GDP. If a 90-day pilot cannot demonstrate an annualized run-rate near the policy's 25% volume uplift and contracted/observed saving, defer ESP and retain CZE only. Conversely, verified demand above base with stable returns strengthens the case for the second site. A real decision should add implementation ramp, discounting, taxes, capacity, service-level effects and working-capital cash timing before final long-term commitment.

4. 90-day implementation and gates

The plan treats the board's approval as staged authorization, not an assumption that both sites are live on day one.

Window	Owner	Work / dependency	Decision gate / measure
Days 0–15	COO + Finance	Confirm CZE/ESP service promise, lane volumes, SKU mix, baseline CZE, design escape codes and capex, release financials	Gate 0: CZE/ESP service promise, lane volumes, SKU mix, baseline CZE, design escape codes and capex, release financials
Days 16–45	Ops + 3PL / Procurement	Run CZE pilot with selected lanes; contract local handling and return	Gate 1: 90% shipped orders and restock targets
Days 46–75	Commercial + CX	Instrument customer promise, quote-to-ship time, service contacts, cost to serve	Gate 2: CZE annualized cost, calibrate ESP de-risk
Days 76–90	Board + COO	Review pilot evidence, updated scenario with actuals, ramp/capacity/cash impact, release ESP only if volume > 20% uplift, shipping > 90%	Gate 3: CZE annualized cost, calibrate ESP de-risk

KPI plan

Weekly by hub: shipped orders, shipped units, gross/net sales, contribution, on-time ship rate, order-to-ship hours, local fulfillment cost per order, saving EUR per unit, return rate by units, restocked recovery rate, open exceptions and FTE utilization. Monthly board dashboard: incremental contribution versus policy low/base, payback progress, capex committed versus budget, SLA, repeat contacts, and cash/refund timing. Each KPI must show numerator, denominator, owner and source system.

Key risks and mitigations: demand uplift risk (stage ESP); FX and price translation risk (reprice and refresh scenario monthly); return inflation (reason codes and restock inspection); duplicate/late data (reconciliation controls); staffing/3PL ramp (cross-train and capacity buffer); and service fragmentation (single customer promise and escalation owner).

5. Quality, limitations and source register

The analysis did not hide known anomalies. They are visible in the workbook's Quality audit tab, analysis/outputs/quality_audit.csv and evidence/source-register-collected.json.

Check	Value	Handling
raw order rows	1328	two extracts plus corrections concatenated
order exact duplicate rows	13	one copy retained
unique order ids after highest revision	1297	highest numeric revision wins
higher-revision order winners	18	replacement row used, not additive
eligible 2025 shipped non-test orders	1254	included even zero-price; cancelled/test/future excluded
raw return rows	264	deduplicated by return_id/revision
return exact duplicate rows	20	one copy retained
higher-revision return winners	1	highest revision used
valid return records	241	eligible linked order and received by 2026-01-31
orphan returns quarantined	1	not matched by guess
returns after cutoff	1	excluded from 2025 accounting
zero-price eligible orders	12	valid shipments included
monthly-country reconciliation	PASS	monthly sums equal country totals

Material limitations

Client exports are synthetic and generated for this exercise; there are no customer interviews or live customer-account claims. The scenario is not a DCF and has no ramp, tax, financing, synergy, capacity or statistical uncertainty model. World Bank population/GDP are contextual only; GDP is current US\$, not PPP or constant-price real income. ECB reference rates are official archived observations used as analytical translation rates. One orphan return (ABSENT) was quarantined and one return received 1 February 2026 was excluded from the cutoff. Returns were assigned to original sale month for accounting, so the contribution view is not a cash-flow timeline.

Source register / evidence links

[S1] Client source room index: <http://127.0.0.1:49683/> (downloaded inputs are preserved under evidence/raw/). [S2] World Bank population archive: https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/SP.POP.TOTL?date=2022:2024&format=json&per_page=1000. [S3] World Bank GDP per capita archive: https://api.worldbank.org/v2/country/DEU;FRA;NLD;POL;CZE;ESP/indicator/NY.GDP.PCAP.CD?date=2022:2024&format=json&per_page=1000. [S4] ECB historical reference rates: <https://www.ecb.europa.eu/stats/eurofxref/eurofxref-hist.zip>. Original URLs, retrieval vintage and SHA-256 hashes are recorded in evidence/source-register-collected.json; all figures in this report are generated from saved local inputs.